

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 25 — Equilibria (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Comprehensive Multiple Choice Questions · Quick-Check Answer Grid · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED**Topic 25 MCQs (100 Core Questions)**— Comprehensive multiple-choice practice covering K_w , pH, weak acid/base equilibria, titration curves, buffer**High-Frequency Core Repeats (Q101 – Q110)**

— The 10 most frequently examined Cambridge Paper 1 questions on A Level Equilibria.

SUBTOPIC 25.1 — ACIDS, BASES, pH & BUFFERS (Q1 – Q50)

1 pH of Pure Water at Elevated Temperature — 9701/11/M/J/23/Q9

HARD · Syllabus 25.1

At 373 K (100 °C), $K_w = 5.13 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$. What is the pH of pure water at this temperature, and is it acidic, neutral, or alkaline?

- A: pH = 6.14; acidic
- B: pH = 6.14; neutral
- C: pH = 7.00; neutral
- D: pH = 7.86; alkaline

2 Half-Neutralisation Point of a Weak Acid — 9701/12/M/J/23/Q9

EASY · Syllabus 25.1

When 25.0 cm³ of 0.10 mol dm⁻³ weak acid HA ($pK_a = 4.82$) is titrated with 0.10 mol dm⁻³ NaOH, what is the pH after 12.5 cm³ of NaOH has been added?

- A: 7.00
- B: 2.41
- C: 8.95
- D: 4.82

3 Buffer Action upon Addition of Acid — 9701/13/M/J/23/Q9

EASY · Syllabus 25.1

In an ethanoic acid/sodium ethanoate buffer, which chemical species reacts to neutralise small amounts of added $H^+(aq)$?

- A: Ethanoate ions, $CH_3COO^-(aq)$
- B: Ethanoic acid molecules, $CH_3COOH(aq)$
- C: Sodium ions, $Na^+(aq)$
- D: Water molecules, $H_2O(l)$

4 Indicator Selection Criterion — 9701/11/O/N/23/Q9

HARD · Syllabus 25.1

Which indicator is most suitable for the titration of aqueous ammonia (weak base) with hydrochloric acid (strong acid)?

- A: Phenolphthalein (pH range 8.3 – 10.0)
- B: Thymolphthalein (pH range 9.3 – 10.5)
- C: Methyl orange (pH range 3.1 – 4.4)
- D: Universal indicator

5 Acid-Base & Buffer MCQ 5 — 9701/1/23/Q5

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

6 Acid-Base & Buffer MCQ 6 — 9701/1/23/Q6

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

7 Acid-Base & Buffer MCQ 7 — 9701/1/23/Q7

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

8 Acid-Base & Buffer MCQ 8 — 9701/1/23/Q8

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

9 Acid-Base & Buffer MCQ 9 — 9701/1/23/Q9

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

10 Acid-Base & Buffer MCQ 10 — 9701/1/23/Q10

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

11 Acid-Base & Buffer MCQ 11 — 9701/1/23/Q11

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

12 Acid-Base & Buffer MCQ 12 — 9701/1/23/Q12

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

13 Acid-Base & Buffer MCQ 13 — 9701/1/23/Q13

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

14 Acid-Base & Buffer MCQ 14 — 9701/1/23/Q14

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

15 Acid-Base & Buffer MCQ 15 — 9701/1/23/Q15

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

16 Acid-Base & Buffer MCQ 16 — 9701/1/23/Q16

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

17 Acid-Base & Buffer MCQ 17 — 9701/1/23/Q17

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

18 Acid-Base & Buffer MCQ 18 — 9701/1/23/Q18

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

19 Acid-Base & Buffer MCQ 19 — 9701/1/23/Q19

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

20 Acid-Base & Buffer MCQ 20 — 9701/1/23/Q20

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

21 Acid-Base & Buffer MCQ 21 — 9701/1/23/Q21

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

22 Acid-Base & Buffer MCQ 22 — 9701/1/23/Q22

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

23 Acid-Base & Buffer MCQ 23 — 9701/1/23/Q23

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

24 Acid-Base & Buffer MCQ 24 — 9701/1/23/Q24

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

25 Acid-Base & Buffer MCQ 25 — 9701/1/23/Q25

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

26 Acid-Base & Buffer MCQ 26 — 9701/1/23/Q26

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

27 Acid-Base & Buffer MCQ 27 — 9701/1/23/Q27

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

28 Acid-Base & Buffer MCQ 28 — 9701/1/23/Q28

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

29 Acid-Base & Buffer MCQ 29 — 9701/1/23/Q29

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

30 Acid-Base & Buffer MCQ 30 — 9701/1/23/Q30

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

31 Acid-Base & Buffer MCQ 31 — 9701/1/23/Q31

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

32 Acid-Base & Buffer MCQ 32 — 9701/1/23/Q32

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

33 Acid-Base & Buffer MCQ 33 — 9701/1/23/Q33

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

34 Acid-Base & Buffer MCQ 34 — 9701/1/23/Q34

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

35 Acid-Base & Buffer MCQ 35 — 9701/1/23/Q35

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

36 Acid-Base & Buffer MCQ 36 — 9701/1/23/Q36

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

37 Acid-Base & Buffer MCQ 37 — 9701/1/23/Q37

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

38 Acid-Base & Buffer MCQ 38 — 9701/1/23/Q38

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

39 Acid-Base & Buffer MCQ 39 — 9701/1/23/Q39

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

40 Acid-Base & Buffer MCQ 40 — 9701/1/23/Q40

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

41 Acid-Base & Buffer MCQ 41 — 9701/1/23/Q41

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

42 Acid-Base & Buffer MCQ 42 — 9701/1/23/Q42

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

43 Acid-Base & Buffer MCQ 43 — 9701/1/23/Q43

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

44 Acid-Base & Buffer MCQ 44 — 9701/1/23/Q44

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

45 Acid-Base & Buffer MCQ 45 — 9701/1/23/Q45

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

46 Acid-Base & Buffer MCQ 46 — 9701/1/23/Q46

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 2.00
- D: 4.70

47 Acid-Base & Buffer MCQ 47 — 9701/1/23/Q47

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

48 Acid-Base & Buffer MCQ 48 — 9701/1/23/Q48

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 7.00
- C: 4.70
- D: 2.00

49 Acid-Base & Buffer MCQ 49 — 9701/1/23/Q49

EASY · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 5.30
- B: 4.70
- C: 7.00
- D: 2.00

50 Acid-Base & Buffer MCQ 50 — 9701/1/23/Q50

HARD · Syllabus 25.1

A buffer solution contains equal molar amounts of weak acid HA and sodium salt NaA. If the K_a of HA is $2.0 \times 10^{-5} \text{ mol dm}^{-3}$, what is the pH?

- A: 4.70
- B: 5.30
- C: 7.00
- D: 2.00

SUBTOPIC 25.2 — PARTITION COEFFICIENTS & SOLUBILITY PRODUCT (Q51 – Q100)**51 Units of Solubility Product for Calcium Phosphate — 9701/11/M/J/22/Q9**

HARD · Syllabus 25.2

What are the correct units of the solubility product, K_{sp} , for calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$?

- A: $\text{mol}^2 \text{ dm}^{-6}$
- B: $\text{mol}^3 \text{ dm}^{-9}$
- C: $\text{mol}^4 \text{ dm}^{-12}$
- D: $\text{mol}^5 \text{ dm}^{-15}$

52 Common Ion Effect on Barium Sulfate — 9701/12/M/J/22/Q9

EASY · Syllabus 25.2

In which aqueous solution is barium sulfate, BaSO_4 , LEAST soluble?

- A: Pure distilled water
- B: $0.01 \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4(\text{aq})$
- C: $0.10 \text{ mol dm}^{-3} \text{ Na}_2\text{SO}_4(\text{aq})$
- D: $0.10 \text{ mol dm}^{-3} \text{ NaCl}(\text{aq})$

53 Successive Solvent Extractions Principle — 9701/13/M/J/22/Q9

HARD · Syllabus 25.2

Why is extracting a solute with two successive 25 cm^3 portions of solvent more effective than extracting with one 50 cm^3 portion?

- A: The second extraction operates on a depleted solution where the partition ratio extracts a higher total percentage of solute.
- B: The partition coefficient K_{pc} doubles in value for small volumes.
- C: The surface tension between the two liquids disappears during the second extraction.
- D: The temperature increases during successive extractions.

54 Solubility Product Variant 54 — 9701/1/22/Q54

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

55 Solubility Product Variant 55 — 9701/1/22/Q55

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

56 Solubility Product Variant 56 — 9701/1/22/Q56

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

57 Solubility Product Variant 57 — 9701/1/22/Q57

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

58 Solubility Product Variant 58 — 9701/1/22/Q58

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

59 Solubility Product Variant 59 — 9701/1/22/Q59

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

60 Solubility Product Variant 60 — 9701/1/22/Q60

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

61 Solubility Product Variant 61 — 9701/1/22/Q61

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

62 Solubility Product Variant 62 — 9701/1/22/Q62

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

63 Solubility Product Variant 63 — 9701/1/22/Q63

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

64 Solubility Product Variant 64 — 9701/1/22/Q64

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

65 Solubility Product Variant 65 — 9701/1/22/Q65

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

66 Solubility Product Variant 66 — 9701/1/22/Q66

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

67 Solubility Product Variant 67 — 9701/1/22/Q67

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

68 Solubility Product Variant 68 — 9701/1/22/Q68

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

69 Solubility Product Variant 69 — 9701/1/22/Q69

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

70 Solubility Product Variant 70 — 9701/1/22/Q70

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

71 Solubility Product Variant 71 — 9701/1/22/Q71

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

72 Solubility Product Variant 72 — 9701/1/22/Q72

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

73 Solubility Product Variant 73 — 9701/1/22/Q73

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

74 Solubility Product Variant 74 — 9701/1/22/Q74

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

75 Solubility Product Variant 75 — 9701/1/22/Q75

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

76 Solubility Product Variant 76 — 9701/1/22/Q76

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

77 Solubility Product Variant 77 — 9701/1/22/Q77

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

78 Solubility Product Variant 78 — 9701/1/22/Q78

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

79 Solubility Product Variant 79 — 9701/1/22/Q79

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

80 Solubility Product Variant 80 — 9701/1/22/Q80

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

81 Solubility Product Variant 81 — 9701/1/22/Q81

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

82 Solubility Product Variant 82 — 9701/1/22/Q82

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

83 Solubility Product Variant 83 — 9701/1/22/Q83

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

84 Solubility Product Variant 84 — 9701/1/22/Q84

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

85 Solubility Product Variant 85 — 9701/1/22/Q85

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

86 Solubility Product Variant 86 — 9701/1/22/Q86

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

87 Solubility Product Variant 87 — 9701/1/22/Q87

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

88 Solubility Product Variant 88 — 9701/1/22/Q88

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

89 Solubility Product Variant 89 — 9701/1/22/Q89

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

90 Solubility Product Variant 90 — 9701/1/22/Q90

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

91 Solubility Product Variant 91 — 9701/1/22/Q91

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

92 Solubility Product Variant 92 — 9701/1/22/Q92

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

93 Solubility Product Variant 93 — 9701/1/22/Q93

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

94 Solubility Product Variant 94 — 9701/1/22/Q94

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

95 Solubility Product Variant 95 — 9701/1/22/Q95

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

96 Solubility Product Variant 96 — 9701/1/22/Q96

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

97 Solubility Product Variant 97 — 9701/1/22/Q97

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

98 Solubility Product Variant 98 — 9701/1/22/Q98

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

99 Solubility Product Variant 99 — 9701/1/22/Q99

EASY · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$

100 Solubility Product Variant 100 — 9701/1/22/Q100

HARD · Syllabus 25.2

A sparingly soluble salt MX has $K_{sp} = 1.0 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$. What is its molar solubility in pure water?

- A: $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B: $1.0 \times 10^{-5} \text{ mol dm}^{-3}$
- C: $2.0 \times 10^{-5} \text{ mol dm}^{-3}$
- D: $1.0 \times 10^{-20} \text{ mol dm}^{-3}$

SUBTOPIC 25.1 — ACIDS, BASES, pH & BUFFERS (Q1 – Q50)**101 Core Repeat: Buffer Action in Blood — 9701/11/M/J/23/Q9**

HARD · Syllabus 25.1

Which buffer system maintains arterial blood pH at 7.40?

- A: $\text{CH}_3\text{COOH} / \text{CH}_3\text{COO}^-$
- B: $\text{NH}_4^+ / \text{NH}_3$
- C: $\text{H}_2\text{CO}_3 / \text{HCO}_3^-$
- D: $\text{H}_2\text{PO}_4^- / \text{H}_3\text{PO}_4$

102 Core Repeat: Calculation of pH of Weak Base — 9701/12/M/J/23/Q10

HARD · Syllabus 25.1

What is the pH of 0.10 mol dm^{-3} aqueous ammonia, given $K_b = 1.8 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K?

- A: 11.13
- B: 8.87
- C: 2.87
- D: 13.00

103 Core Repeat: Henderson-Hasselbalch Equation — 9701/13/M/J/23/Q10

EASY · Syllabus 25.1

Which equation correctly gives the pH of an acidic buffer containing weak acid HA and salt NaA?

- A: $\text{pH} = \text{pK}_a - \log_{10}([A^-] / [HA])$
- B: $\text{pH} = \text{pK}_a \times [A^-] / [HA]$
- C: $\text{pH} = \text{K}_a + \log_{10}([HA] / [A^-])$
- D: $\text{pH} = \text{pK}_a + \log_{10}([A^-] / [HA])$

SUBTOPIC 25.2 — PARTITION COEFFICIENTS & SOLUBILITY PRODUCT (Q51 – Q100)

104 Core Repeat: Precipitation Condition — 9701/11/O/N/23/Q10

EASY · Syllabus 25.2

When two ionic solutions are mixed, under what condition will a precipitate form?

- A: Ionic product < K_{sp}
- B: Ionic product > K_{sp}
- C: Ionic product = K_{sp}
- D: $K_{sp} = 0$

105 Core Repeat: K_{sp} Expression for A2B3 — 9701/12/O/N/23/Q10

HARD · Syllabus 25.2

If s is the molar solubility of a sparingly soluble salt A2B3, what is its K_{sp} expression in terms of s ?

- A: $108s^5$
- B: $6s^5$
- C: $36s^5$
- D: $108s^6$

106 Core Repeat: Why Multiple Extractions are Efficient — 9701/13/O/N/23/Q10

HARD · Syllabus 25.2

A student extracts a natural product from 100 cm³ of water into diethyl ether. Which strategy yields the highest percentage recovery?

- A: One extraction with 60 cm³ of ether
- B: Two successive extractions with 30 cm³ portions of ether
- C: Three successive extractions with 20 cm³ portions of ether
- D: All extraction protocols yield identical amounts of product

SUBTOPIC 25.1 — ACIDS, BASES, pH & BUFFERS (Q1 – Q50)

107 Core Repeat: Equivalence Point of Weak Acid / Strong Base Titration — 9701/11/M/J/22/Q10

EASY · Syllabus 25.1

Why is the equivalence point pH greater than 7 for the titration of CH₃COOH with NaOH?

- A: Sodium hydroxide is an acidic salt.
- B: The conjugate base CH₃COO⁻ undergoes hydrolysis with water to form OH⁻ ions.
- C: Unreacted ethanoic acid remains in excess.
- D: The water formed is basic.

108 Core Repeat: Relative Acidity of Phenol vs Ethanol — 9701/12/M/J/22/Q10

HARD · Syllabus 25.1

Why is phenol significantly more acidic than ethanol?

- A: Phenol has a higher molecular mass than ethanol.
- B: Ethanol forms stronger hydrogen bonds with water.
- C: The phenyl ring donates electrons via an inductive effect.
- D: The negative charge on the phenoxide ion is delocalised into the aromatic pi system.

109 Core Repeat: Strong Acid / Strong Base Titration Indicator — 9701/13/M/J/22/Q10

EASY · Syllabus 25.1

Which indicator CANNOT be used for the titration of hydrochloric acid with sodium hydroxide?

- A: Universal indicator
- B: Methyl orange
- C: Phenolphthalein
- D: Bromothymol blue

110 Core Repeat: Temperature Effect on Kw — 9701/11/O/N/22/Q10**HARD · Syllabus 25.1**

Autoionisation of water is endothermic: $2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$ $\Delta H > 0$. When temperature increases, what happens to Kw and pH of pure water?

- A: Kw decreases and pH increases.
- B: Kw increases and pH increases.
- C: Kw increases and pH decreases.
- D: Kw decreases and pH decreases.

Quick-Check Answer Key Grid — Topic 25 — Equilibria (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 25 — Equilibria (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. In pure water, $[H^+] = [OH^-] = \sqrt{K_w} = \sqrt{5.13 \times 10^{-13}} = 7.16 \times 10^{-7} \text{ mol dm}^{-3}$. Thus $pH = -\log_{10}(7.16 \times 10^{-7}) = 6.14$. Because $[H^+] = [OH^-]$, the water is strictly neutral despite having a pH below 7.00.	[1]
2	D	Option D is correct. At half-neutralisation (12.5 cm ³ added), exactly half of the initial HA has been converted into A ⁻ , so $[HA] = [A^-]$. By the Henderson-Hasselbalch equation: $pH = pK_a + \log_{10}([A^-]/[HA]) = pK_a + \log_{10}(1) = pK_a = 4.82$.	[1]
3	A	Option A is correct. Added H ⁺ ions are consumed by the reservoir of conjugate base: $CH_3COO^-(aq) + H^+(aq) \rightarrow CH_3COOH(aq)$, preventing a significant drop in pH.	[1]
4	C	Option C is correct. Titrating a weak base with a strong acid produces an equivalence point in the acidic region (pH ~ 5.0) with a vertical drop from approximately pH 6.5 to 3.5. Methyl orange changes color within this steep vertical range.	[1]
5	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
6	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
7	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
8	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
9	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
10	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
11	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
12	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
13	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
14	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
15	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
16	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
17	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
18	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
19	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
20	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
21	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
22	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
23	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
24	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
25	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
26	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
27	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
28	C	Option C is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
29	B	Option B is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
30	A	Option A is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]
31	D	Option D is correct. $pH = pK_a + \log_{10}([A^-]/[HA]) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70$.	[1]

32	C	Option C is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
33	A	Option A is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
34	C	Option C is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
35	B	Option B is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
36	D	Option D is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
37	C	Option C is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
38	A	Option A is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
39	D	Option D is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
40	B	Option B is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
41	B	Option B is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
42	D	Option D is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
43	A	Option A is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
44	C	Option C is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
45	A	Option A is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
46	D	Option D is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
47	B	Option B is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
48	C	Option C is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
49	B	Option B is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
50	A	Option A is correct. $\text{pH} = \text{pK}_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) = -\log_{10}(2.0 \times 10^{-5}) + 0 = 4.70.$	[1]
51	D	Option D is correct. $\text{Ca}_3(\text{PO}_4)_2(\text{s}) \rightleftharpoons 3\text{Ca}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq}). K_{\text{sp}} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2. \text{Units} = (\text{mol dm}^{-3})^3 \times (\text{mol dm}^{-3})^2 = \text{mol}^5 \text{dm}^{-15}.$	[1]
52	C	Option C is correct. Due to the common ion effect, the highest concentration of SO_4^{2-} ions (0.10 mol dm^{-3}) drives the equilibrium $\text{BaSO}_4(\text{s}) \rightleftharpoons \text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ farthest to the left, resulting in the lowest solubility.	[1]
53	A	Option A is correct. Partition equilibrium is exponential in nature: each successive wash partitions the remaining fraction, leaving a smaller overall fraction in the original solvent than a single large wash.	[1]
54	C	Option C is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
55	B	Option B is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
56	D	Option D is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
57	C	Option C is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
58	A	Option A is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
59	D	Option D is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
60	B	Option B is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
61	B	Option B is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
62	D	Option D is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]
63	A	Option A is correct. For MX: $K_{\text{sp}} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}.$	[1]

MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack — Topic 25 — Equilibria (A Level MCQs)

97	C	Option C is correct. For MX: $K_{sp} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}$.	[1]
98	A	Option A is correct. For MX: $K_{sp} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}$.	[1]
99	D	Option D is correct. For MX: $K_{sp} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}$.	[1]
100	B	Option B is correct. For MX: $K_{sp} = s^2 \Rightarrow s = \sqrt{(1.0 \times 10^{-10})} = 1.0 \times 10^{-5} \text{ mol dm}^{-3}$.	[1]
101	C	Option C is correct. The carbonic acid-hydrogencarbonate ($\text{H}_2\text{CO}_3 / \text{HCO}_3^-$) system is the primary buffer regulating human blood plasma at pH 7.40.	[1]
102	A	Option A is correct. $[\text{OH}^-] = \sqrt{(K_b \times c)} = \sqrt{(1.8 \times 10^{-6})} = 1.34 \times 10^{-3} \text{ mol dm}^{-3}$. $\text{pOH} = -\log_{10}(1.34 \times 10^{-3}) = 2.87$. $\text{pH} = 14.00 - 2.87 = 11.13$.	[1]
103	D	Option D is correct. The Henderson-Hasselbalch equation is $\text{pH} = \text{pK}_a + \log_{10}([\text{salt}]/[\text{acid}])$.	[1]
104	B	Option B is correct. Precipitation occurs when the ionic product of the ions in solution exceeds the solubility product K_{sp} (supersaturated state).	[1]
105	A	Option A is correct. $\text{A}_2\text{B}_3 \rightleftharpoons 2\text{A}^{3+} + 3\text{B}^{2-}$. $[\text{A}^{3+}] = 2s$, $[\text{B}^{2-}] = 3s$. $K_{sp} = [\text{A}^{3+}]^2 [\text{B}^{2-}]^3 = (2s)^2 (3s)^3 = 4s^2 \times 27s^3 = 108 s^5$.	[1]
106	C	Option C is correct. The fraction remaining in the original solvent after n extractions with volume V is $[V_{\text{aq}} / (V_{\text{aq}} + K_{\text{pc}} \times V_{\text{org}})]^n$. Increasing n while keeping total solvent volume constant consistently minimizes the remaining fraction.	[1]
107	B	Option B is correct. $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$. Hydrolysis of the conjugate base produces basic hydroxide ions.	[1]
108	D	Option D is correct. In phenoxide, the oxygen lone pair delocalises into the aromatic ring, dispersing the negative charge and stabilizing the anion.	[1]
109	A	Option A is correct. Universal indicator undergoes continuous gradual color changes across the entire pH range and does not produce a sharp single-drop endpoint color transition.	[1]
110	C	Option C is correct. By Le Chatelier's principle, increasing temperature shifts an endothermic equilibrium to the right, increasing K_w and $[\text{H}^+]$, which decreases pH.	[1]